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Providing Actionable Information to Avoid Induced Seismicity Associated with Produced Water Injection Using Machine Learning

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Abstract

Produced water associated with unconventional shale oil and gas production is an ever-growing challenge. A relatively cost-effective option to the disposal of this produced water is re-injection into the subsurface. While there is a minimal upfront cost to this option, there is the known risk of triggering earthquakes. Oklahoma and West Texas have recently experienced large, 5.6 magnitude quakes that can be attributed to produced water disposal. There is little guidance available to operators and governing bodies, however, to manage subsurface water disposal in a safe manner. This paper provides such a tool using a machine learning-based approach to quantitatively predict the risk of induced seismicity in the future given a chosen injection location and injection volume.

We analyze two large publicly available datasets documenting earthquake activity and produced water injection trends and injection locations for both Oklahoma as well as the Permian Basin in Texas. We exploit time-dependent relationships between the earthquake activity and injection trends to provide a causal relationship between location/volume of injection and the probability of inducing an earthquake. We also document how this approach extends to predicting possible earthquake magnitudes.

The time-dependent relationships between earthquake activity and injection characteristics allow us to predict the likelihood of an earthquake two years in advance with a high degree of fidelity. The predictive ability of our approach degrades marginally beyond a two-year period but provides for a quantitative measure of risk associated with injection to allow for planning of injection locations and sequence of injection over the near-term future.

Prior approaches to predict earthquake activity or intensity have generally relied on a purely data driven approach without considering the temporal effect of injection. Using a time-series approach, we can create a causal model relevant to specific time windows and therefore we do not overestimate, nor do we underestimate the potential risks. Critically, we analyze both the temporal and spatial components of earthquake activity induced by produced water injection, we present a robust and validated predictive model.

Introduction

Induced seismicity resulting from produced water injection and saltwater disposal (SWD) is a well-documented geological hazard with multiple studies attempting to identify the primary mechanisms driving these seismic events (Keranen et al. 2013, 2014; McClure et al. 2016; Walsh and Zoback, 2016; Hincks et al. 2018; Deng et al. 2020; Hicks et al. 2021; Hennings et al. 2024, OGS 2024). The state of Oklahoma provides one of the more well-studied regions for injection-induced seismicity. While historical seismic activity in the state was reported to be low through the 2000s (Frohlich 2009, McNamara et al. 2015), there has been a rapid increase in reported activity during the period of from 2009 to 2017, coinciding with rapid increases in saltwater disposal (Ellsworth, 2013; Keranen et al. 2013; Frohlich et al. 2014, OGS, 2024).

The most significant events include the 2011 Mw 5.6 and 2016 Mw 5.8 earthquakes near Prague, both widely attributed to years of high-volume injection into Arbuckle Group formations (Keranen et al. 2013). Walsh and Zoback (2015) demonstrated a clear correlation between the dramatic increase in SWD volumes and increased seismic activity, while adjacent areas with low disposal volumes experienced far fewer earthquakes. Several other papers have reported strong correlations between injection volume, depth, and the occurrence of induced seismicity (Keranen et al. 2015; Weingarten et al. 2015; Goebel et al. 2017; Hincks et al. 2018).

The Permian Basin in west Texas is another critical area where induced seismicity has recently become a significant problem since the 2020s. Similar to Oklahoma, historical seismic activity in the region was reported to be low but has substantially increased over the recent years (TexNet, 2025).

Given the risks posed by the induced seismic activity, regulatory agencies have implemented various mitigation strategies. The Oklahoma Corporation Commission has developed a "traffic light" protocol system that provides a structured response framework, that manages injection locations in tandem with reported seismic activity. The Texas Railroad Commission has established seismic response areas and implemented several restrictive measures such as restriction of allowable injection rates to approximately 10,000 barrels per day and limits on surface pressure calculated to be less than 0.25 per foot.

As early as 2013, Llenos and Michael (2013) provided an initial study into the observed increases in seismic activity in Oklahoma. Subsequently, Keranen et al. (2013–2014), Weingarten et al. (2015), and Walsh and Zoback (2015) also reported strong correlations between injection volumes and seismicity rates. Walsh and Zoback (2015) also report a 9-month lag between injection and observed seismicity. McClure et al. (2016) also implemented a statistical analysis to query the association between the increased seismicity and produced water injection in Oklahoma and confirmed this relationship with high statistical confidence. In another work, Langenbruch and Zoback (2016), developed a statistical model linking cumulative injection volumes to the frequency of earthquakes with magnitude $\geq M$ 3.0. Their model incorporated threshold injection rates and temporal lag effects and predicted that reducing injection rates would lead to a gradual decrease in earthquake frequency over time. Hincks et al. (2018) employed Bayesian network modeling to identify the strongest predictors of seismic moment release and determined that injection depth relative to crystalline basement represents the most significant control factor. The study demonstrated that constraining injection depths to 200–500 meters above the basement could reduce seismic moment release. Johann et al. (2018) developed a physics-based model that combines pore pressure diffusion and poroelastic coupling mechanisms and successfully reproduced observed spatiotemporal patterns of induced seismicity.

There have also been several attempts to provide predictive capabilities using machine learning tools. An ML prediction model for short-term earthquake forecasting was developed by Xiong et al. (2021) using satellite data while Oliver and Misra (2024) use causal analyses to understand the link between injection and seismicity. They also report that their analysis did not indicate any link between seismic magnitude and injection volume/location.

In this work, we explore spatiotemporal patterns in earthquake activity and wastewater injection and provide predictive capability for seismic risk in response to earthquake location, injection volumes and

pressures. By combining concepts developed in [Walsh and Zoback \(2015\)](#) with a data-driven spatial analysis, we are able to identify low-risk locations for injection, and quantify seismic risk in both the Midland basin and Oklahoma that provides can lead to actionable insights for managing produced water injection.

Data Description

In this work, we conduct two separate analyses: one for Oklahoma and the other for the Permian Basin.

Permian Basin Data

This study also focused on the Permian Basin region within the geographic boundary of approximately 31°N to 35°N latitude and 101°W to 105°W longitude. Data was collected from the publicly available Texas Seismological Network and Seismology Research (TexNet). The dataset includes well injection parameters such as injection volume (BBLs) and average injection pressure (PSIG), along with surface coordinates (latitude and longitude). Earthquake magnitudes were used between mid-2018 and mid-2025, and injection data from 2019 to 2024 were incorporated with a 12-month lag to account for the delayed seismic response following injection operations. For model development, the input features— injection volume, injection pressure, and location coordinates—were separated from the target variable, which was defined as earthquake density. Following data cleaning and preprocessing to eliminate outliers, a total of 2,192 data points were retained for analysis. The dataset was subsequently partitioned into training and testing subsets using an 80:20 split, and all input features were normalized using min-max scaling to ensure consistency in model training.

[Fig. 1](#) shows the earthquake locations in the Permian Basin with each datapoint colored by the seismic magnitude. In this work, we restricted our analyses to the Midland Basin adjacent to the cities of Midland and Odessa. The analyzed earthquake locations are shown in [Fig. 2](#).

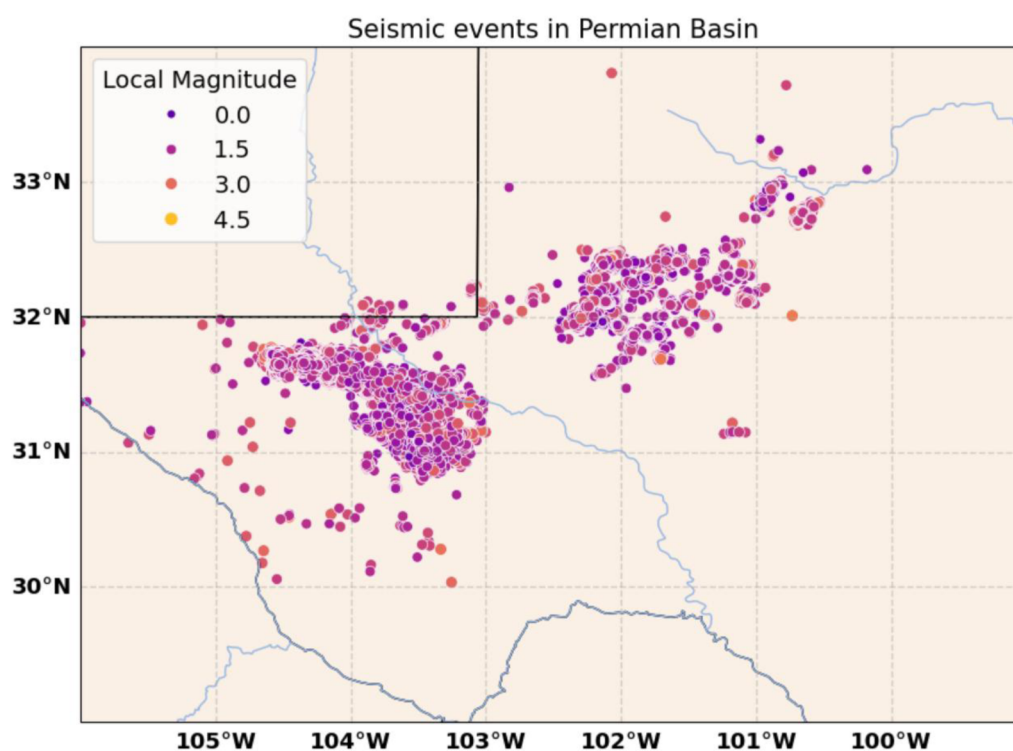


Figure 1—Earthquake locations up to 2025 in the Permian Basin of West Texas.

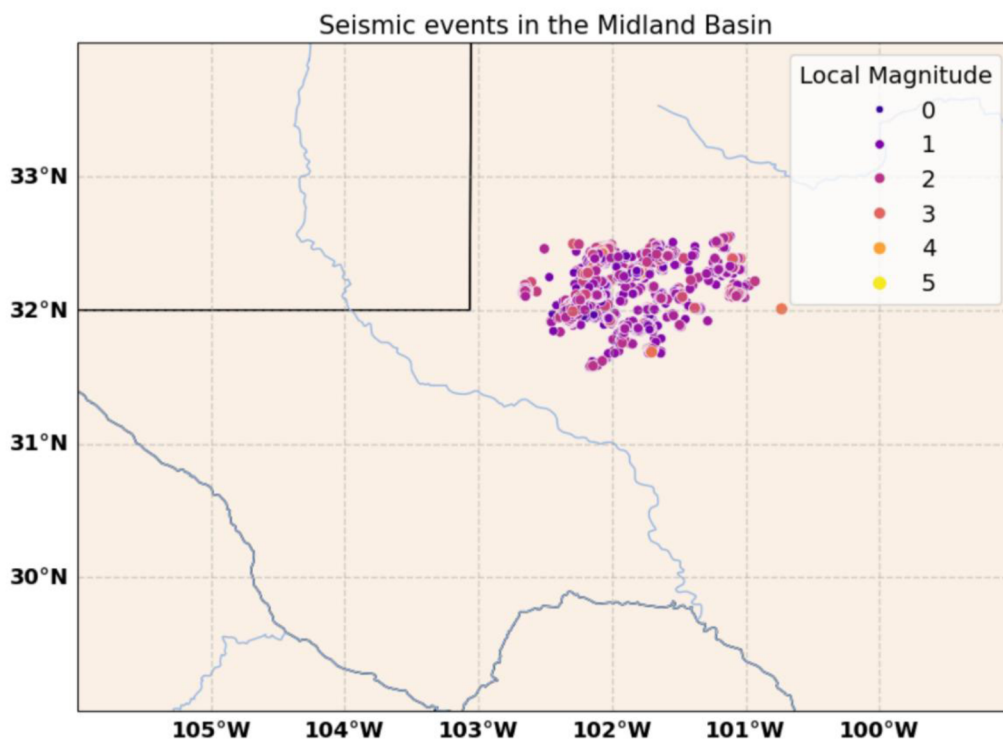


Figure 2—Earthquake locations up to 2025 in the Midland Basin in west Texas adjacent to the cities of Midland and Odessa.

Data from Oklahoma

The study utilized earthquake and wastewater injection data from Oklahoma Corporation Commission (OCC) and the United States Geological Survey (USGS), focusing on the geographic region bounded by 33.5°N to 37°N latitude and 96°W to 102°W longitude. Earthquake magnitudes were collected from mid-2013 to mid-2016, while injection data spanned 2011 to 2014, accounting for a 29-month time lag between injection activities and seismic response—a critical factor in assessing induced seismicity. Earthquake events and injection volumes were aggregated on a monthly basis, and monthly average injection pressures were calculated to standardize the temporal resolution of the dataset. Following data cleaning and preprocessing to remove outliers and incomplete records, the study analyzed 384 total data points. The dataset was partitioned into 345 (90%) training samples and 39 (10%) testing samples, ensuring robust statistical evaluation.

Fig. 3 shows the earthquake locations analyzed in Oklahoma. While there are a few other events to the northwest portion of the state in the Oklahoma panhandle, they are few and far apart and a statistical analysis is not feasible. Also important to mention is that there are over five decades of a history of produced water injection in Oklahoma; however, the more recent developments in the Anadarko Basin has led to larger volumes of produced water and consequently, significantly larger injection volumes. In this work, we focus on the post-2014 history when seismic activity became more pronounced (OGS, 2025)

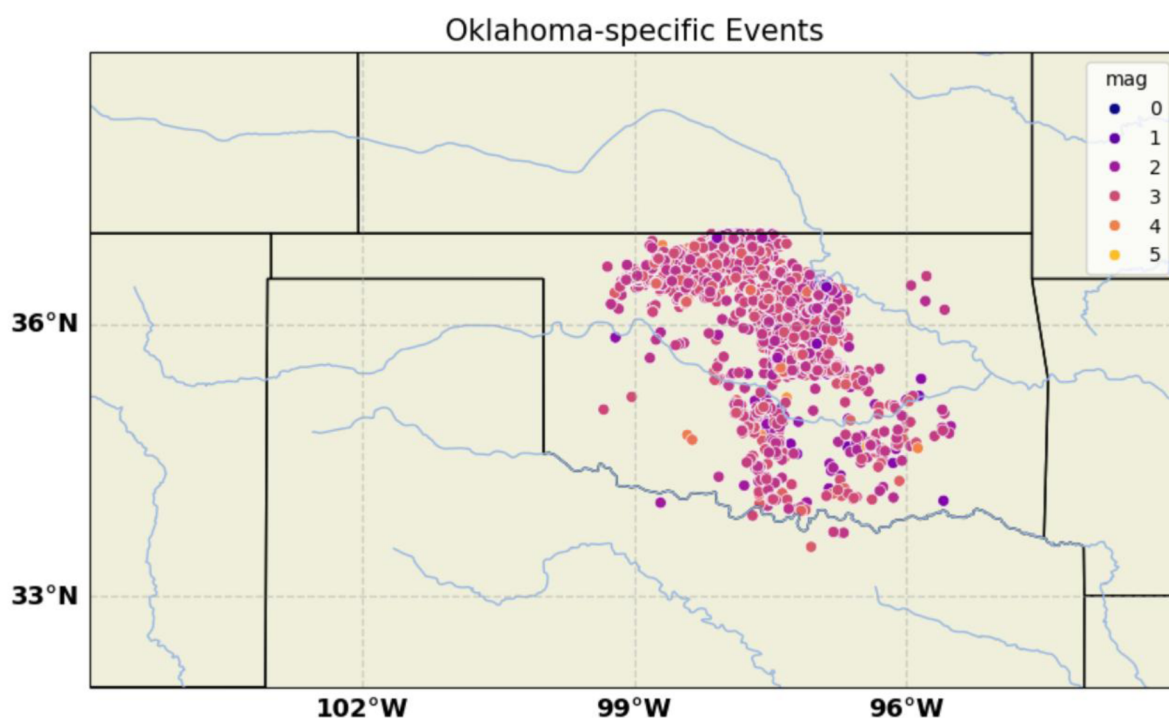


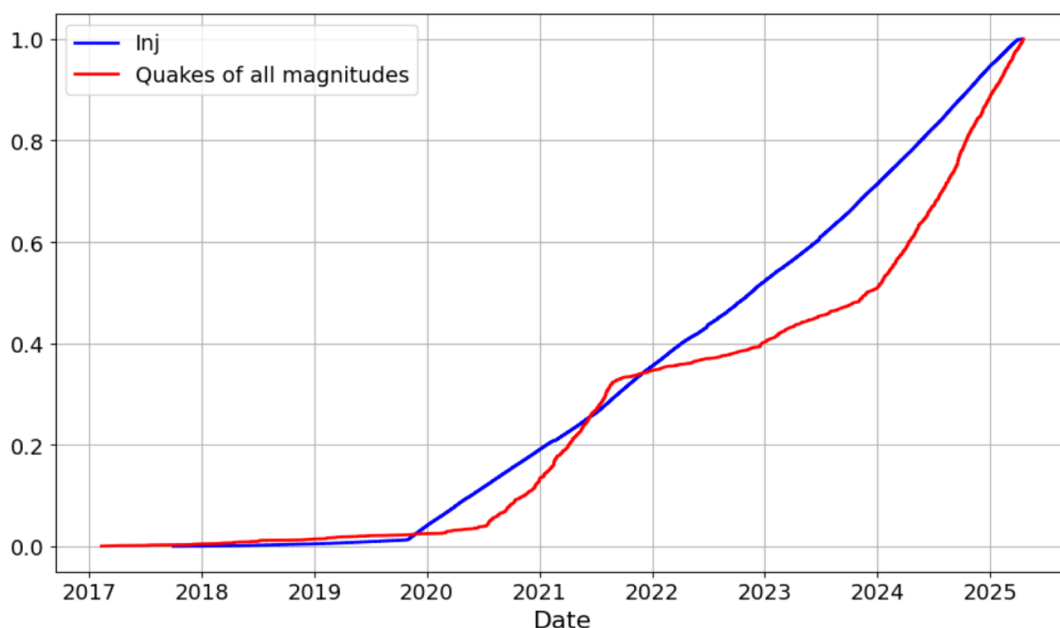
Figure 3—Earthquake locations up to from 2011 to 2019 in Oklahoma. The locations are restricted to between 95.5W and 99.5W which constitutes our area of analysis in Oklahoma.

Methodology For Quantifying Earthquake Risk

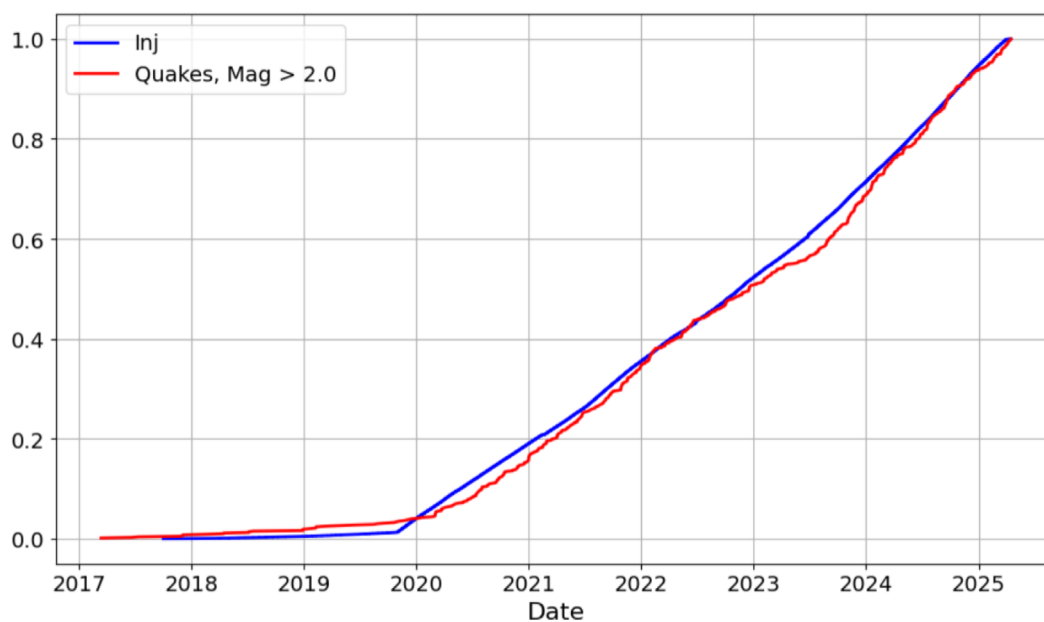
We adopt a similar approach for quantifying the induced seismicity risk in both Oklahoma and the Midland Basin.

Midland Basin Analysis

First, we plot cumulative number of seismic events and cumulative water injection as a function of time to determine a gross lag distance between injection and earthquake activity. The goal is to identify a sudden increase in injection activity and to determine the time interval to an observed uptick in seismic activity. This is illustrated in Fig. 4. Fig. 4a shows the plot of cumulative number of seismic events and produced water injected as a function of time from 2017 to 2025. While the plot shows moderate correlation between the two quantities, there is excellent correspondence when viewing seismic events of magnitude greater than 2.0. In general, humans can feel tremors of magnitude 3.0 and above (USGS, 2023). Viewing this aggregate data, it appears that there is almost an immediate seismic response to produced water injection. It is important to note that this observation is made on aggregate data, with little to no spatial information or detailed temporal analysis.



(a)



(b)

Figure 4—Plots showing the correlation between cumulative number of seismic events and cumulative water injection over time. The scales are normalized from 0.0 to 1.0. In the Midland Basin, (a) shows moderate correlation between the number of seismic events and injection volumes, while (b) shows a remarkable correlation between the number of seismic events and cumulative volume of injection. No time shift is necessary indicating an immediate seismic response to produced water injection.

The next step in the process is to then identify regions within the study area that are more prone to earthquakes. We accomplish this by simply constructing a density map of earthquake locations in the study area. In one dimension, a density plot is simply the probability distribution function. Our plot in Fig. 5 is the two-dimensional equivalent. However, we do not normalize the density contours. This is unnecessary for our workflow because, as we demonstrate later, we are interested only in the relative magnitudes of the earthquake density over the study area. Fig. 5 also shows the injection well locations overlaid on the plot.

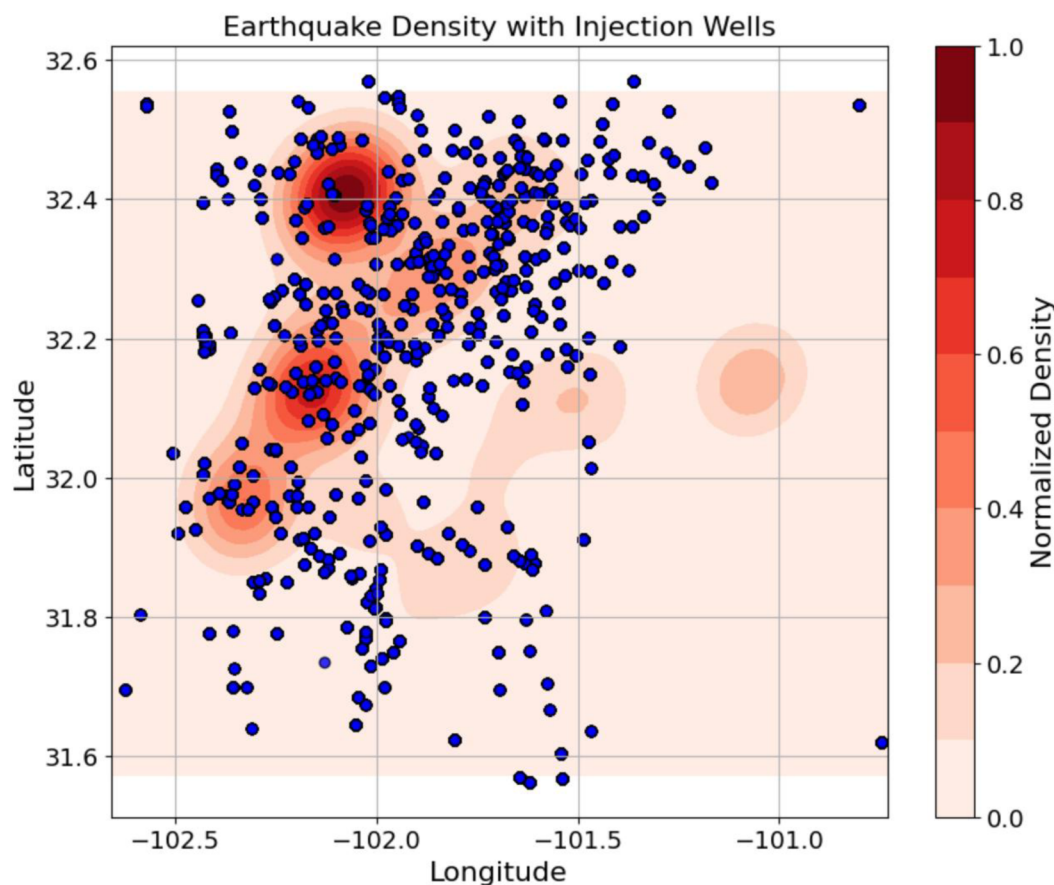
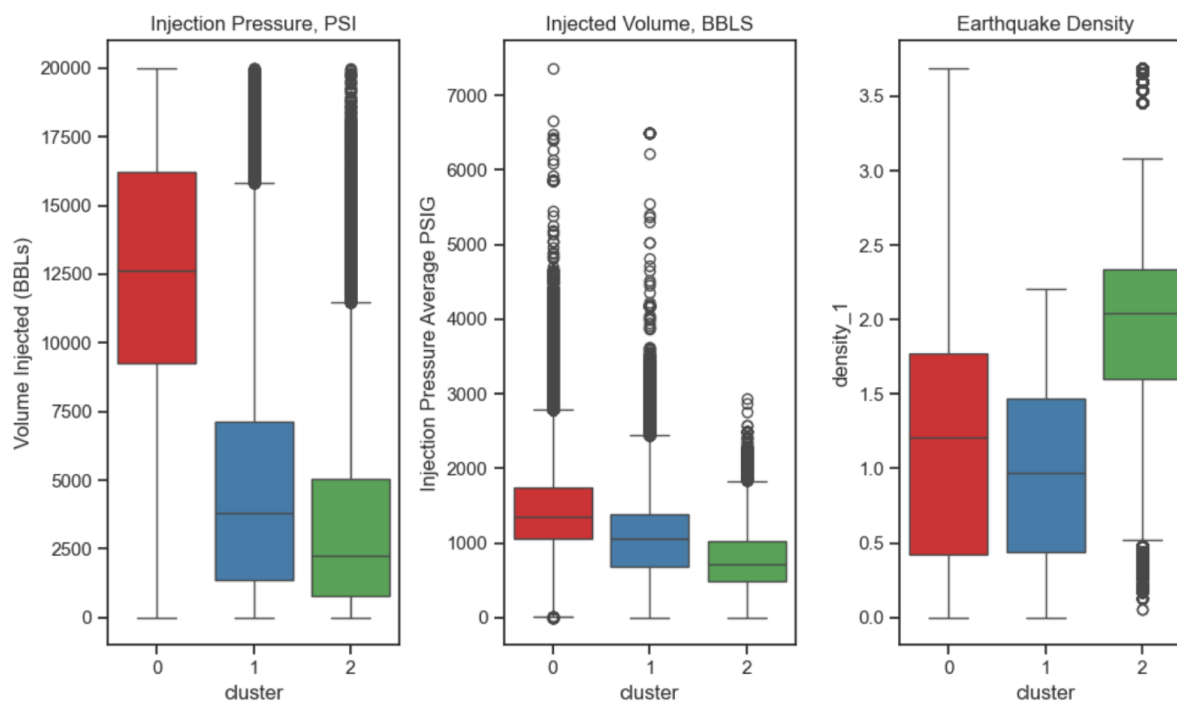


Figure 5—Plot showing the earthquake density contours with the injection wells overlaid. The absolute contour values represent the number of earthquakes that have occurred in the vicinity of a specific location. For this study, we are not interested in this number; instead, the number is simply used to demarcate regions that are more prone to magnitude 2.0 and above earthquakes.

Consequently, the plot shows regions within the study area that are more prone to earthquakes of magnitude 2.0 and above. The next step in our analysis involves clustering the wells within the study area using location (latitude, longitude), computed earthquake density (such as in Fig. 5), injection pressure and cumulative injection volume. The rationale for this approach is to identify the linkage between injection pressure, computer earthquake density, injection volumes and location as done in Fig. 6.

Fig. 6a shows the characteristics of each well cluster. Cluster 2 shows the highest tendency towards seismic activity with a high adjacent computer earthquake density. Unfortunately, this is also associated with the lowest cumulative injection volumes and injection pressures. It is likely that there are underlying geologic causes for this observation. On the other hand, Cluster 1 wells take in almost equivalent volumes of water at similar pressures and are associated with very low seismic risk. The locations of the wells belonging to each cluster are shown in Fig. 6b. It is important to mention that the analyses presented so far is predicated on the availability of high-fidelity earthquake and injection data. Another caveat to note is that the earthquake density map is continually evolving. Nevertheless, its contemporary form is indicative of the likelihood of induced seismicity in the vicinity of an injection well.



(a)

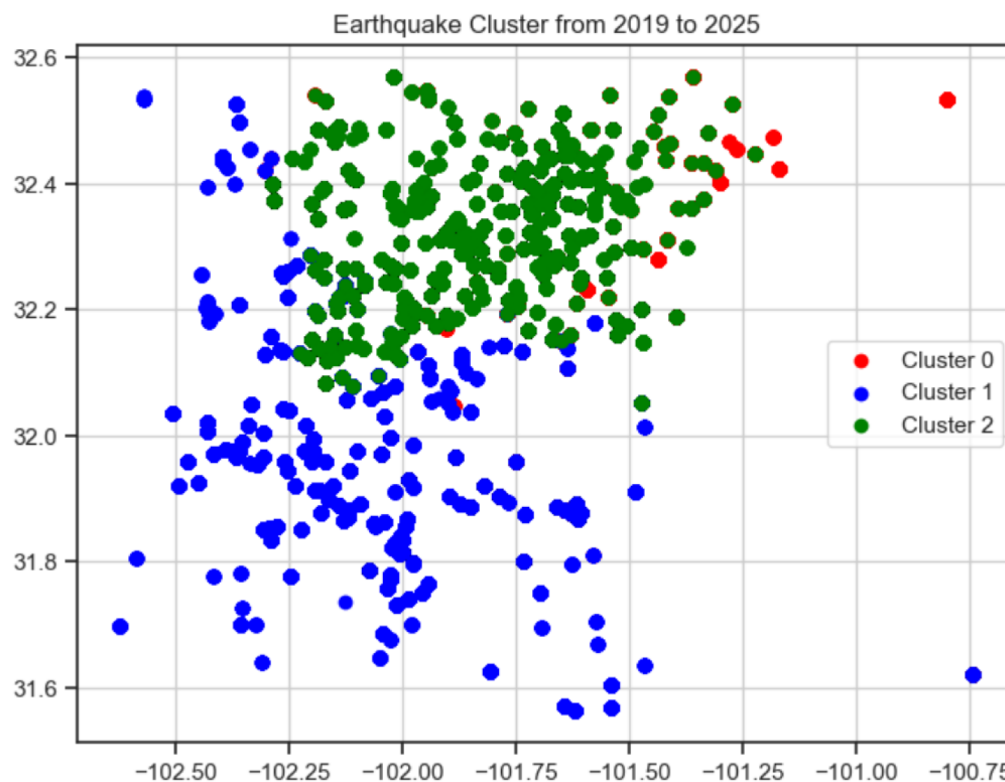


Figure 6—Plot showing the results of clustering using well locations, computed earthquake density, cumulative injection volume and average injection pressure as inputs. The figure on the top in (a) shows the properties of the three clusters. All three clusters vary moderately in terms of the injection pressures. Cluster 2 is associated with the highest earthquake density and are more tremor prone and is associated with the lowest injection volumes and lowest injection pressures. Wells in Cluster 1 are less tremor prone and are associated with lower injection volumes. The figure on the bottom in (b) shows the spatial location of the three clusters.

The next step is to predict the risk of injection-induced earthquakes for any given location in the study area. Injection risk is quantified through the computed earthquake density shown in Fig. 5. The explanatory variables chosen for this regression model are latitude and longitude, cumulative injection volume, and injection pressure while the predicted, dependent variable is earthquake density. We use a random forest regressor with 5-fold crossvalidation for hyperparameter tuning to select the best model. As can be seen in Fig. 7 plotting the true density on the x-axis versus the predicted density on the y-axis, there is excellent correspondence on the test data. This indicates that for any given current or proposed well location, we can predict the likelihood of earthquakes in the vicinity provided estimates of the injection pressure and cumulative volume is available. Again, we would like to point out there is a caveat to our approach. The approach does not consider the dissipation of pore pressure gradually if injection were to be more episodic and gradual, instead of the larger injection rates that are typical.

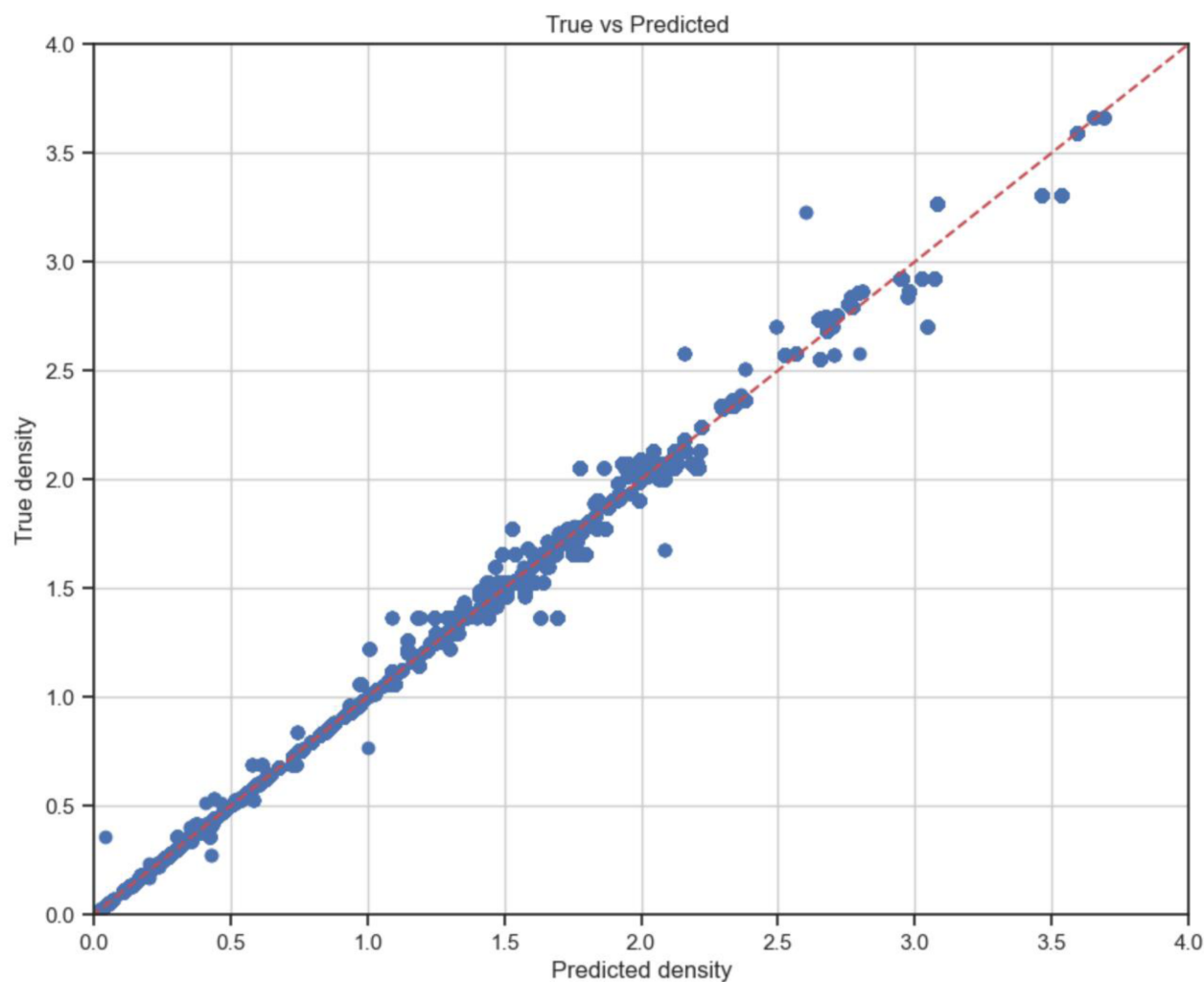


Figure 7—Plot showing true versus predicted earthquake density or the propensity for earthquakes in the vicinity of any location. The inputs to the regression model are current or proposed well location, cumulative injection volume and average injection pressure. The R2 value on the test dataset was 0.99.

An alternative approach is to predict whether an earthquake is likely or not at a specific location given the injection volume and injection pressure. We achieve this by training a classifier on the well locations, cumulative injection volumes and injection pressures. The predicted label is whether an earthquake will occur or not. The label is 1 when the earthquake density in the vicinity of an injection well is more than 1.0 and 0 when the earthquake density adjacent to an injection well is less than 1.0. Fig. 8 shows the results of the

trained classifier applied to the test data demonstrating that the approach developed in this paper is sufficient to predict earthquake risk reliably. We now show a similar approach applied to the Oklahoma dataset.

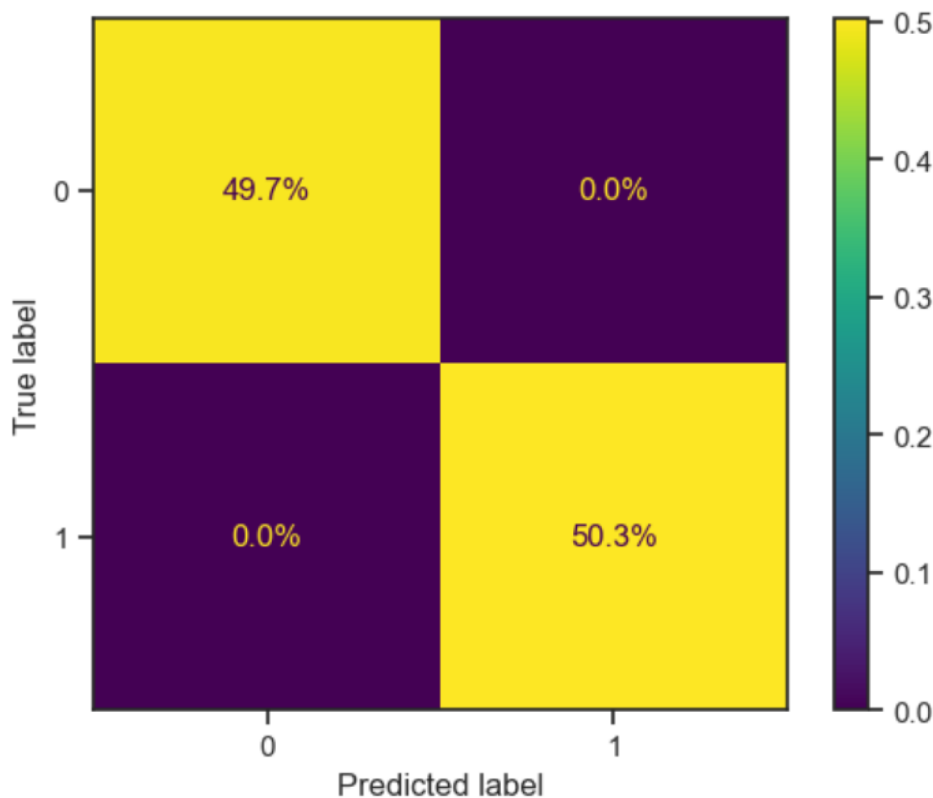
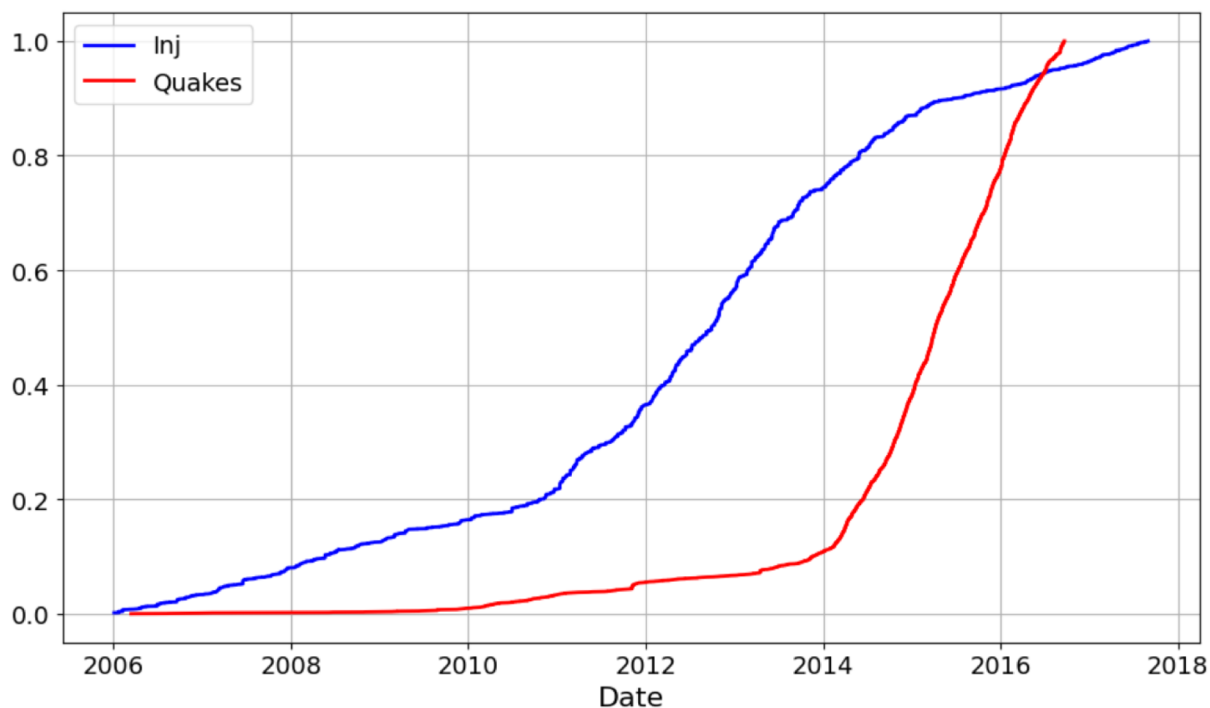


Figure 8—Plot showing the confusion matrix associated with a trained classifier. The label 1 is associated with locations where the earthquake density is more than a specified threshold and is 0 when the density is lower. The accuracy of prediction is over 99%, indicating the reliability of our approach.

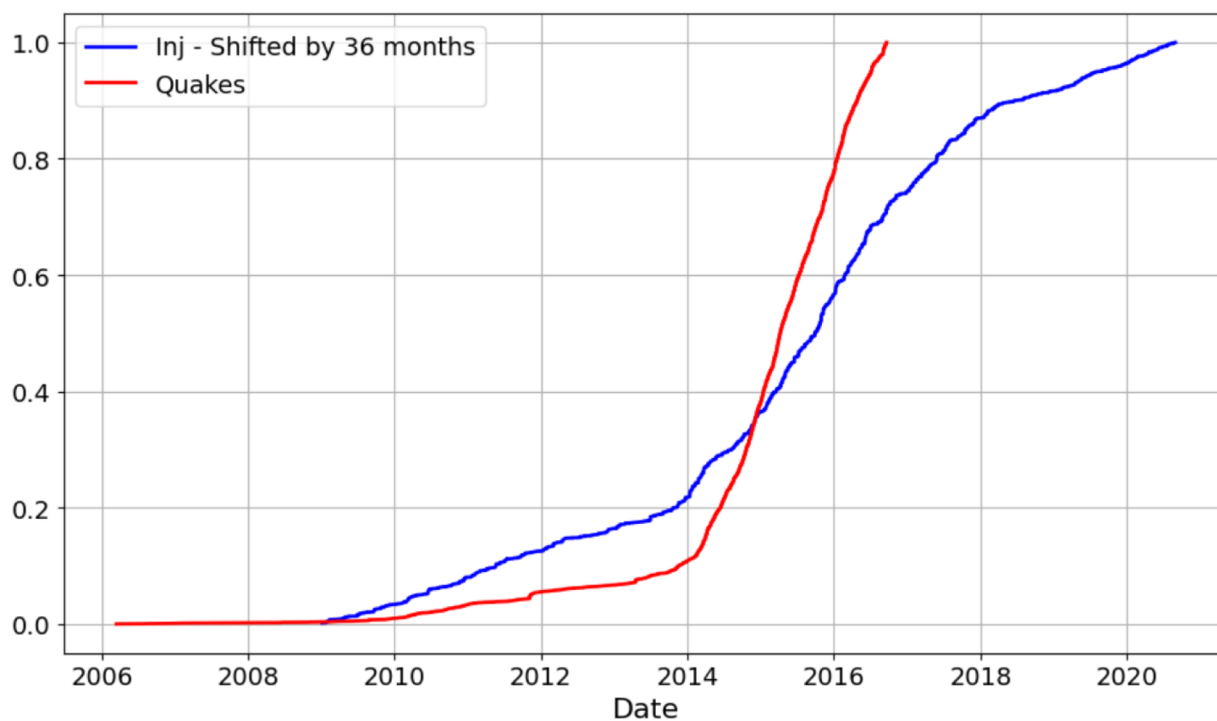
Oklahoma Analysis

As mentioned earlier, the data from Oklahoma spans a timeframe from 2006 to 2019. Fig. 9a shows the plot of cumulative water injection and the cumulative number of seismic events in Oklahoma. The data has been normalized to illustrate the similarity in the overall trends. The induced seismicity is delayed relative to the injection and when shifted by 36 months as seen in Fig. 9b, we observe an initial period of quiescence, leading to a moderate increase in injection volumes and induced seismicity. Subsequently, as injection rates increase, denoted by the sharp increase in cumulative injection volumes around 2014, we also see a corresponding sharp increase in (the shifted) seismic activity. With this data, on an aggregate basis, seismic activity is delayed by approximately 36 months relative to injection in Oklahoma. This contrasts with the observations in Fig. 4 for the Midland basin, where synchronous seismic activity and injection are observed.

There are likely several operational and geologic reasons for this observation, including the nearby presence of critically stressed faults, larger volumes of injection, higher pressures and others.



(a)



(b)

Figure 9—The plot in (a) shows normalized cumulative injection and normalized seismic activity. In (b), when the seismic activity is shifted earlier by 36 months, the smaller scale trends in injection and seismic activity show several similarities beginning with a period of no activity, followed by moderate activity and then a steep increase around 2014.

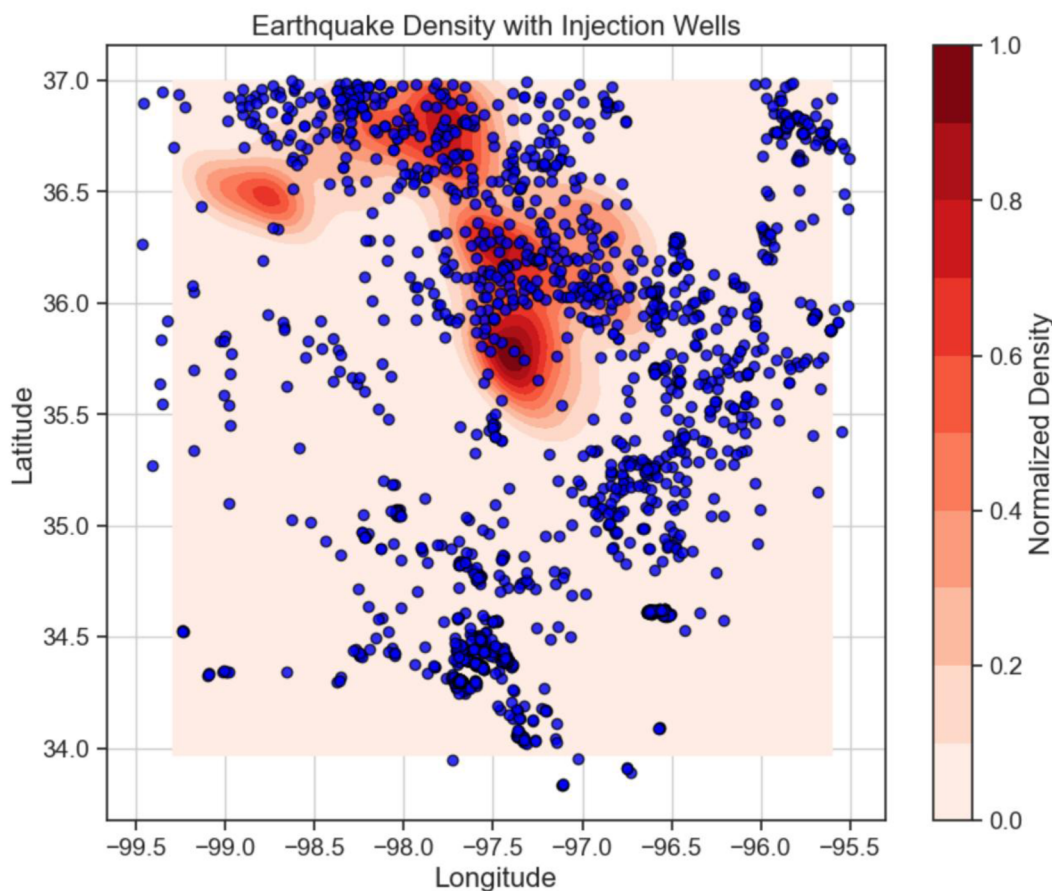
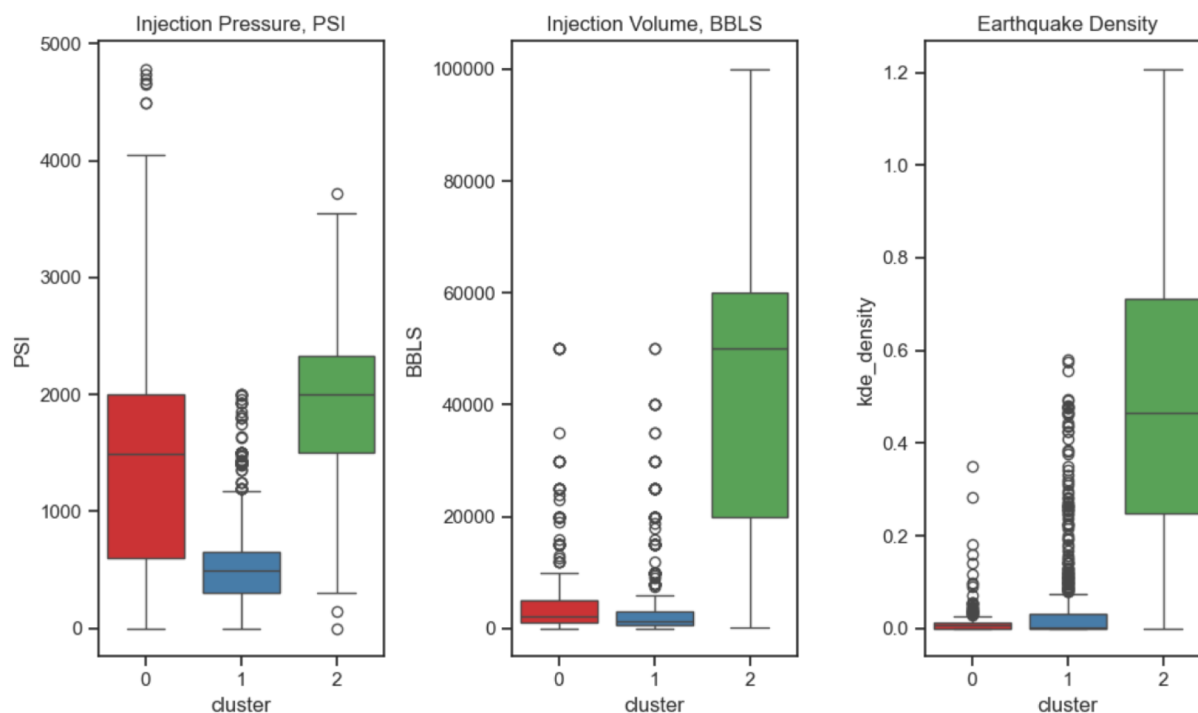


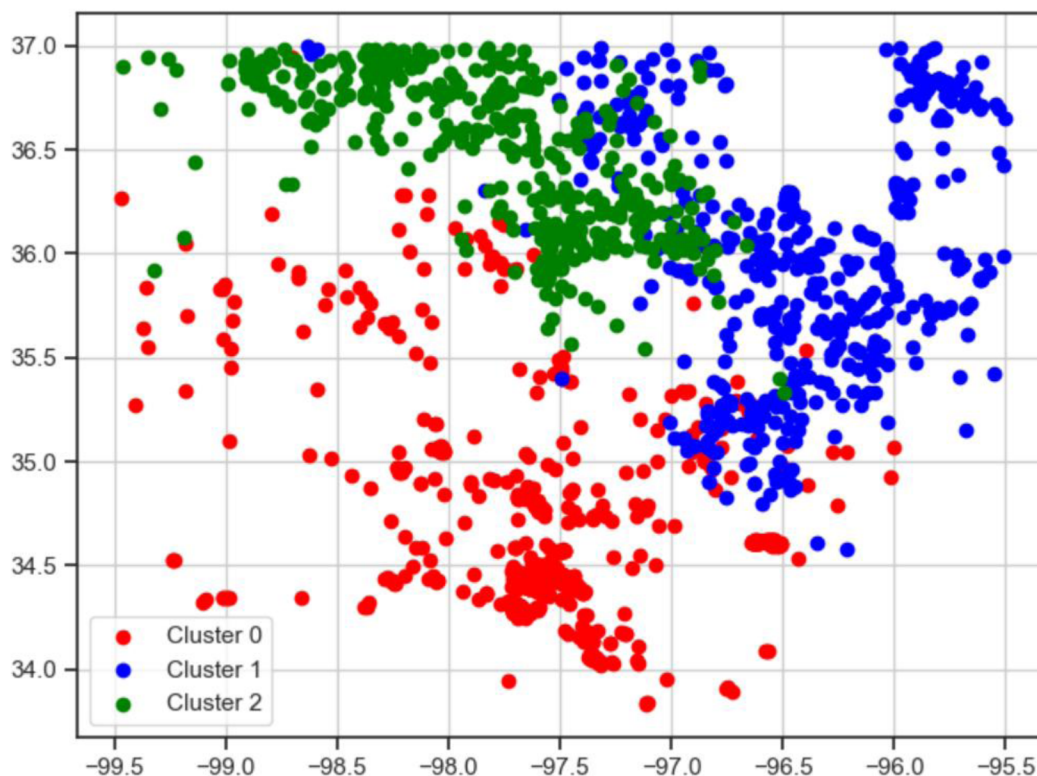
Figure 10—Plot showing earthquake density at 2018 with the location of injection wells in the Oklahoma region.

In a manner analogous to the workflow for the Midland basin, we then generate a 2D density plot of earthquake activity and quantify the densities associated with each injection well in the region as shown in Fig. 10. Subsequently, using location, injection pressure and cumulative injection volume at 2018 as inputs, we cluster the wells into three groups as shown in Fig. 11a. Cluster 2 is associated with the highest induced seismicity, the highest injection pressures and injection volumes. Cluster 0 wells are also subject to high injection volumes but lower pressures and do not experience high levels of induced seismicity in the vicinity. Fig. 11b shows the locations of the wells in a latitude-longitude map of the region.

Analogous to what was shown in Fig. 8 and 9, the next step is to predict the risk of injection-induced earthquakes for any given location in the study area. Injection risk is quantified through the computed earthquake density shown in Fig. 5. The explanatory variables chosen for this regression model are latitude and longitude, cumulative injection volume, and injection pressure while the predicted, dependent variable is earthquake density. We use a random forest regressor with 5-fold crossvalidation for hyperparameter tuning to select the best model. As can be seen in Fig. 12 plotting the true density on the x-axis versus the predicted density on the y-axis, there is excellent correspondence on the test data. This indicates that for any given current or proposed well location, we can predict the likelihood of earthquakes in the vicinity provided estimates of the injection pressure and cumulative volume is available.



(a)



(b)

Figure 11—Plot showing the results of clustering using well locations, computed earthquake density, cumulative injection volume and average injection pressure as inputs. The figure on the top in (a) shows the properties of the three clusters. Cluster 2 is associated with the highest earthquake density and is associated with the highest injection volumes and highest injection pressures. Wells in Cluster 1 are less tremor prone and are associated with higher injection volumes at significantly lower pressures. The figure on the bottom in (b) shows the spatial location of the three clusters.

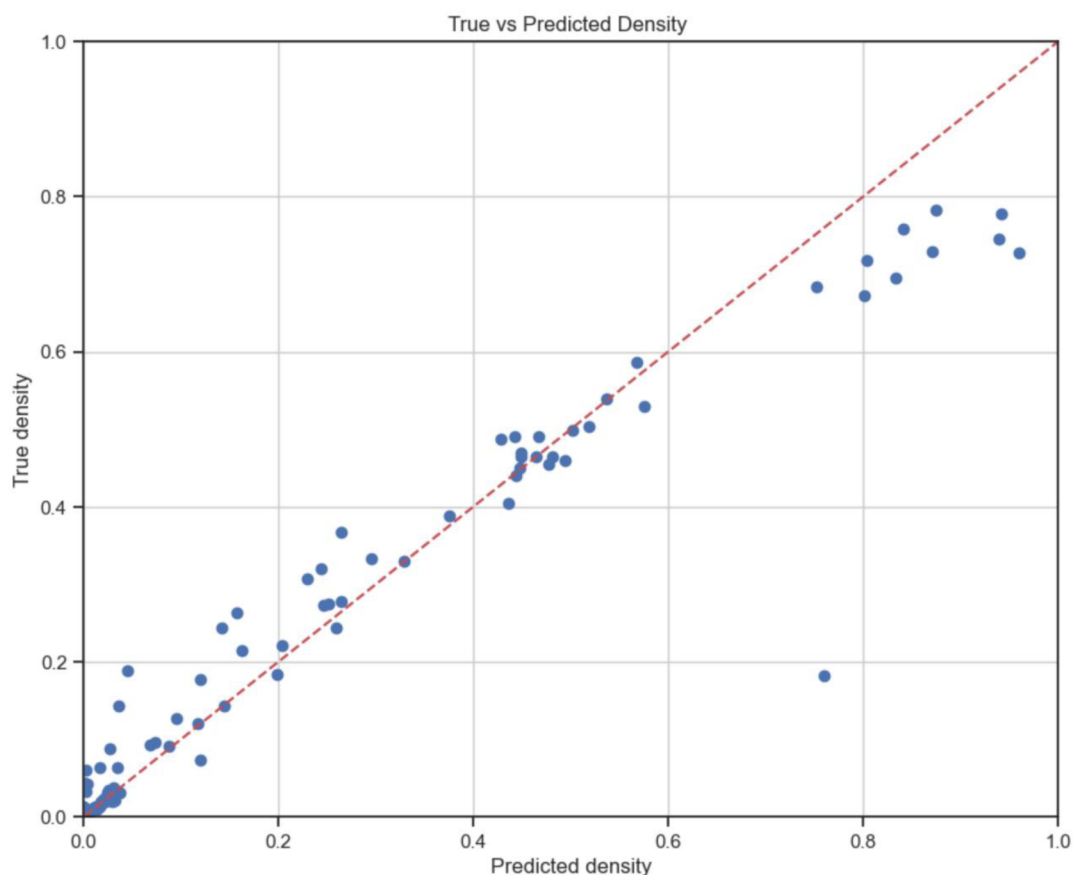


Figure 12—Plot showing true versus predicted earthquake density or the propensity for earthquakes in the vicinity of any location. The inputs to the regression model are current or proposed well location, cumulative injection volume and average injection pressure. The R2 value on the test dataset was 0.99.

We can also view the prediction of induced seismicity as a classification problem and label regions with a high earthquake density as distinct from regions of lower earthquake density. Using latitude and longitude, cumulative injection volume, and injection pressure as predictors, we can classify wells as low risk or high risk. In this work, we again use a random forest classifier with 5-fold crossvalidation and the confusion matrix related to the test data is shown in [Fig. 13](#).

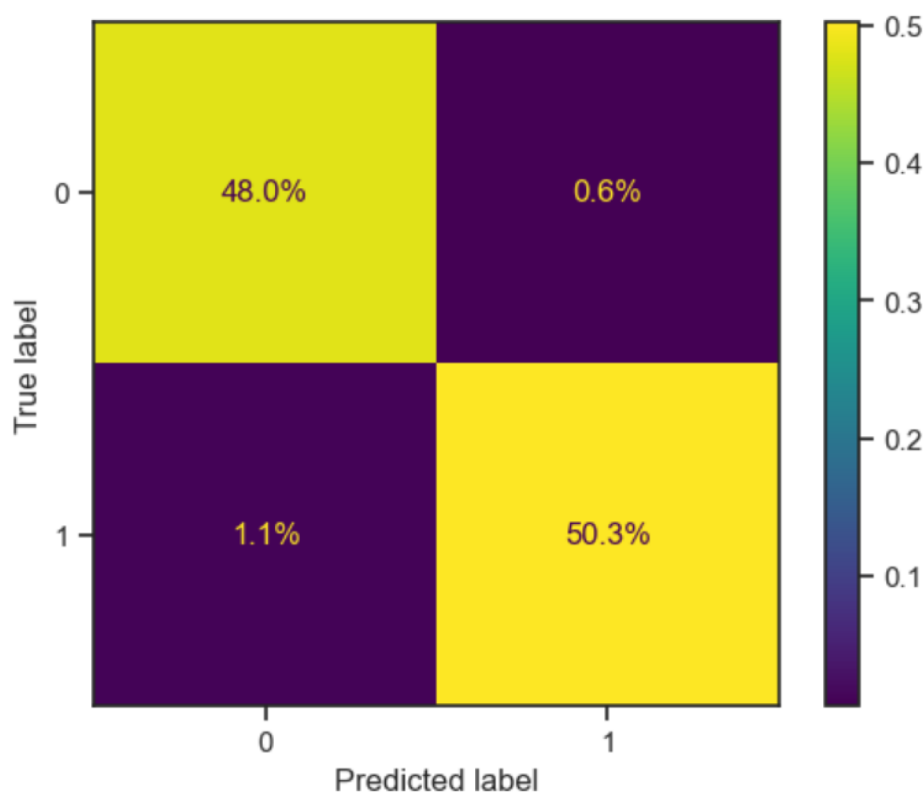


Figure 13—Plot showing the confusion matrix associated with a trained classifier. The label 1 is associated with locations where the earthquake density is more than a specified threshold and is 0 when the density is lower. The accuracy of prediction is over 99%, indicating the reliability of our approach.

Relevance and Practical Applicability of Proposed Approach

Predicting the risk of induced seismicity occurring because of produced water injection continues to be a topic of high interest, especially in the Permian basin and in Oklahoma. With the wealth of data available, there are several studies that have progressively raised awareness of the issue and also provided a scientific rationale for the observed induced seismicity. We are fortunate with the abundance of data that allowed us to adopt a purely data-driven approach. As documented in two separate case studies, there is great promise in our approach. However, there are a few caveats that need to be mentioned before the approach can be generalized to other locations.

First is the availability of large volumes of data without which our approach simply cannot proceed. Extracting temporal trends such as in Figs. 4b and 9b would simply not be possible without the data. Secondly, we make a few assumptions in our analysis. We are predicting seismic risk using historical data. As with the use of any historical data, the assumption is that the processes governing the behavior of the system are likely to continue and govern future behavior. This is true of production forecasting using decline curves, for instance, although there are several instances where the assumptions do not hold. Well interference, change of artificial lift, or completion re-design all necessitate a closer look at the production forecasts. This is true of our workflow as well. When injection is managed by the regulatory authorities, there may or may not be a link to historical injection/seismic activity. However, we did not test this in our work.

Conclusions

We demonstrate a machine learning approach that combines spatial and temporal statistical analyses to provide an estimate of induced seismicity risk in response to produced water injection. We apply the

approach to two case studied: one centered around the Midland basin in west Texas and the other in Oklahoma. The following conclusions can be drawn from this work:

1. In both case studies, the temporal delay in seismicity in response to injection was unique: 36 months in the case of Oklahoma while in west Texas, both occurred synchronously.
2. There are geologic and operational underpinnings to induced seismicity. The clustered wells in both Oklahoma and west Texas were different in location, injection pressures and injection volumes.
3. The machine learning approach we adopt to predict earthquake prone areas is characterized by a low false positive and low false negative rate and consequently, not only highly accurate, but provides predictive capabilities that are meaningful and actionable.

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